Birch Hollow Water Treatment Plant

1 Design Basis

The Birch Hollow Water Treatment Plant is sized for a design flow of 13.1 MGD (average day) and a peak flow of 45.8 MGD on the maximum day, serving a population of 76,127 persons. Washwater recycle pump P-101C returns 779 gpm of washwater to the head of the plant. Washwater recycle pump P-101C has a total dynamic head of 78.5 ft.

Raw water turbidity reaches 14.5 NTU (95th percentile), measured over the last five years. Finished water turbidity shall not exceed 0.13 NTU in 95% of monthly samples.

2 Treatment Process

The rapid mix basin provides a detention time of 24 s at design flow, with a velocity gradient of 957 1/s. Flocculation follows in 3 stages, for a total detention time of 30.3 min at design flow.

There are 4 sedimentation basins, each 147 ft long and 33 ft wide, with a side water depth of 15.3 ft; the surface loading rate is 0.82 gpm/ft² at design flow. Sludge pump P-101B moves 1,336 gpm of sludge from the basins. Sludge pump P-101B has a total dynamic head of 60.1 ft.

Filtration uses 11 dual-media filters of 836 ft² each. With one filter out of service, the filtration rate is 3.2 gpm/ft². Backwash pump P-101A delivers 3,395 gpm to the filters. Backwash pump P-101A has a total dynamic head of 67.9 ft.

Table 1. Raw water pumps

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)
P-201A	6,261	112.5	287
P-201B	4,423	62.6	140
P-201C	8,562	115.1	141

3 Chemical Feed

Alum is dosed at 31.6 mg/L on the average day and up to 53.3 mg/L on the maximum day. Chlorine is fed at 2.2 mg/L ahead of the clearwell.

4 Disinfection Options

Two options were compared for primary disinfection.

Table 2. Disinfection options

	Option A: UV	Option B: chlorine contact
Capital cost ($M)	5.04	2.12
Annual energy use (MWh/yr)	813	n/a

Contact time at peak flow (min)	n/a	51